

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 03

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Waste products that are the raw materials for other processes	1
01.2	• Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable) • The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing CO₂	2
01.3	• Incineration uses high temperatures / fuel and releases pollutants; nutrients are not returned as usable raw materials • Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3.	
02.1	<i>Any 4 from:</i> • Zooxanthellae are photosynthetic algae living inside the polyp cells • The algae absorb CO₂ and excretory wastes from the polyp for photosynthesis and growth (efficient nutrient recycling) • They supply the polyp with oxygen and glucose - over 95% of its food energy • In return the algae gain a protected habitat and nutrients they could not absorb from the nutrient-deficient water • Polyps also feed on plankton captured by cilia and nematocysts	4
02.2	• Corals deposit a skeleton of calcium carbonate (calcite) • Each carbonate molecule requires the absorption of a molecule of carbon dioxide, so reefs are a carbon sink	2

Question	Marking guidance	Marks
02.3	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Importance: cover 1% of ocean area but support over a quarter of marine species; fisheries (fish, lobsters, conch); medicines (AZT, Ara-C, dolastatin, secosteroids, bryostatin, trabectedin, bone grafts); climate control as carbon sink; erosion protection of low-lying islands; tourism income • Threats: physical damage (anchors, divers, litter, coral extraction), collection of ornaments (CITES Appendix II), sedimentation from rivers and coastal development, pollution (oil, nutrients causing algal shading, sewage, pesticides, sunscreen), ocean acidification, bleaching from warming, overfishing (giant triton / crown of thorns starfish), introduced species (red lionfish), loss of mangroves and seagrass nurseries • Conservation: marine protected areas / national parks (Great Barrier Reef, Belize, Chagos), mooring buoys, diver education, No Take Zones which can increase catches elsewhere, sustainable ecotourism and fishing, protection of mangroves and seagrass, erosion and pollution control, planning control of ports and resorts • Fishing controls: closed seasons, minimum and maximum catchable sizes, bans on dynamite, trawling and small-mesh nets • Evaluation: local measures cannot control global threats (warming, acidification) which need international action; enforcement in remote areas is difficult; conflicts with local livelihoods; large vs small NTZs • Justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	
03.1	<i>Any 2 from:</i> • Hedgerows • Hay meadows • Grazed moorland / heathland • Chalk grassland	2
03.2	• Drainage produces more aerobic soils so wetland plants and the animals they support cannot survive • Fertilisers increase growth of tall species such as grasses which out-compete smaller plants such as wildflowers	2
03.3	• The introduced (non-indigenous) predator may eat other species as well as the pest	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 328-329; AQA spec 3.5.1.3.	

Question	Marking guidance	Marks
04.1	- By-catch = $5.9 + 1.9 = 7.8$ t; total = 8.4 t $7.8 / 8.4 \times 100 = 92.9\%$ (accept 93%)	2
04.2	The nets have a very small mesh size so few non-target animals can escape	1
04.3	- They breed at an old age (skate about 11 years; Greenland shark 100-150 years) and produce few young - They may reach a catchable size before they are sexually mature, so fishing can remove the whole breeding population before it has bred - Populations recover slowly, unlike r-selected species (herring, mackerel) which breed young and produce many young; they are caught as by-catch in demersal trawls so fishing continues even though they have no commercial value	3
04.4	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) - Impacts: population decline of k-selected species, by-catch (immature target fish, over-quota fish, non-commercial species; drift nets - whales, dolphins, turtles; longlines - albatrosses; pair trawls - porpoises), ghost fishing, seabed damage by demersal trawls, coral and seagrass damage, dynamite fishing, food-web effects - Catch quotas: limit weight landed; work well for single-species pelagic shoals but in mixed fisheries over-quota fish are discarded dead so the quota fails to protect them - Gear design: mesh size and square-mesh escape panels let small fish escape (diagonal mesh closes under drag); turtle exclusion devices; acoustic pingers for dolphins; curved hooks, decoys, sinkers and night setting to protect albatrosses; biodegradable / radio-tracked traps to reduce ghost fishing - Restricted effort: limits on boat size, engine power and days at sea; decommissioning; restricted methods - drift-net bans, demersal trawling bans on sensitive seabeds (Rockall Bank, Hatton Bank) - Evaluation: enforcement at sea is difficult; some methods reduce catch or raise costs; quotas need reliable stock data; effectiveness depends on species (single vs mixed shoals) and on international cooperation as fish migrate; as local stocks decline fishing moves to more distant waters - Justified conclusion weighing effectiveness of different methods	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 341-345; AQA spec 3.5.2.2.	

Question	Marking guidance	Marks
05.1	• SSSI: UK legislation (Wildlife and Countryside Act); best examples of UK flora, fauna, geological features; management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal) • SPA: EU Birds Directive; places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas) • Ramsar: Ramsar Convention (intergovernmental agreement); internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally • MCZ: UK marine legislation (Marine and Coastal Access Act) under the W&CA framework; protects marine habitats / species by restricting activities such as fishing and dredging; NNR: best SSSIs with complete communities	4
05.2	• Legal status: restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage) • Management agreements between owners and statutory bodies to maintain suitable conditions • But it does not protect against external influences such as air pollution or climate change; very few sites are state-owned so protection depends on owner cooperation	3
05.3	• Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues • Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland) • Benefits everyone so society pays through central funding; legislation cannot compel positive management	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.	
06.1	• Moral belief: protecting the environment is the 'right' thing to do regardless of benefit to people • Enlightened self-interest: protecting the environment is done because it causes fewer difficulties (for people) than not doing so	2
06.2	• Total impact = population size x the individual (per capita) impact of each person • Per capita impact rises with affluence / access to energy, so both a larger population and a more affluent population increase the total impact	2
06.3	<i>Any 1 from:</i> • Slash and burn agriculture • Fishing • Discharge of sewage / biodegradable pollutants	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370; AQA spec 3.6.1.	
07.1	• Less dead organic matter / leaf litter so less humus and slower soil formation; fewer nutrients returned • Loss of canopy and litter exposes soil to rain splash and wind • Decomposing roots no longer bind soil so erosion, gullyng and landslides increase (especially on slopes) • Reduced organic matter and biota lower infiltration and water retention	4

Question	Marking guidance	Marks
07.2	• Reduced interception and transpiration so less water returned to the atmosphere and less rainfall downwind • Increased runoff: more flooding after rain, lower flows in dry periods; sedimentation of rivers • Increased albedo of cleared land (reflects more sunlight) • Reduced carbon sequestration and loss of the carbon reservoir (CO₂ released by burning / decay) contributing to climate change	4
07.3	<i>Any 2 from:</i> • Fragmentation of remaining forest into isolated patches with non-viable populations / edge effects • Loss of inter-species relationships (pollinators, seed dispersers) causing further declines • Loss of genetic resources / potential medicines before they are discovered	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 368-369; AQA spec 3.5.3.3.	
08.1	• Species: a group of closely related organisms that naturally interbreed to produce fertile offspring and are reproductively isolated from other groups (Passer domesticus, house sparrow); the only 'real' taxon • Niche: the role a species plays in its habitat - how it uses resources and relates to other species (tawny owl: nocturnal carnivore nesting in tree holes eating small mammals); two species cannot share exactly the same niche in one habitat • Community: all the populations of all species (plants, animals, fungi, bacteria, protists, archaea) living in an area • Biome: a large geographical region with a specific climate and characteristic community, including all areas where it occurs (all tropical rainforests, tundra, savannah)	8
08.2	• An ecosystem is the biotic community plus abiotic factors of one area, relatively self-contained (a particular coral reef) • A biome includes all separate ecosystems with the same climate and community type (all coral reefs); each reef is a different ecosystem within the same biome	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 82-85; AQA spec 3.1.3.2.	
09.1	• Island communities evolved from few colonising species in isolation, often without mammal predators or large herbivores • So indigenous species lack adaptations (eg ground-nesting birds, flightlessness) to survive introduced cats, rats, pigs, goats	2
09.2	• Competitor: grey squirrel (North America) out-competes red squirrel - digests acorns, larger, wins nest sites; rhododendron shades ground and releases toxins preventing native regrowth • Predator: American mink (escaped fur farms) reduced water vole range; cane toads in Australia; cats / rats / pigs on islands eating ground-nesting birds; Nile perch in Lake Victoria ate cichlids to extinction • Pathogen: squirrel pox virus carried by grey squirrels kills red squirrels; crayfish plague carried by signal crayfish kills white-clawed crayfish; Dutch elm disease, ash dieback, sudden oak death imported on plants / soil	6

Question	Marking guidance	Marks
09.3	• Closely related introduced species (sika deer; domestic cats) cross-breed producing fertile hybrids • The natural gene pool is changed / diluted with genes it would not naturally contain, so the pure indigenous species may disappear	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 24-25; AQA spec 3.1.2.2.	
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Water: catchment management, conservation, metering, greywater, re-use of treated effluent, aquifer recharge, sustainable urban drainage - largely cyclical already; limits in arid regions • Minerals: exploiting low-grade ores with low-temperature processes (bioleaching), extending product lifespans, repair / re-use / repurpose / recycle, design for disassembly, reduced packaging; limits: dispersal, alloys, energy of collection • Waste management: repair, re-use, refurbish, recycle, composting and anaerobic digestion returning nutrients, minimal landfill, energy from residual waste; EU targets on packaging, batteries, WEEE • Pollution control: shift from post-production treatment to non-release - fuel desulfurisation not FGD, fuel cells / electric vehicles not combustion, industrial symbiosis (Kalundborg: waste heat, gypsum, fly ash, yeast used by neighbours); limits for legacy pollution • Energy: renewables, storage, energy density technologies, low-temperature manufacturing, CHP; limits of intermittency, transport fuels • Food: natural processes for nutrients (legumes, organic matter), pest control (IPM), soil biota, agri-environment schemes, IMTA aquaculture; tension with maximum yield for a growing population • Land uses supporting ecosystems: urban green infrastructure, gardens, permeable surfaces, habitat in redeveloped docks / lagoons • Judgement: high applicability in some sectors (water, waste), partial in others (minerals, energy for transport); needs policy support	9
10.2	• Locating industries so that the wastes of one (heat, gases, by-products) become the raw materials of another, mimicking natural cycles (eg Kalundborg eco-industrial park)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 375-379; AQA spec 3.6.4.1.	

Question	Marking guidance	Marks
11.1	Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Setting priorities: IUCN Red List criteria, EDGE, endemic, keystone, flagship species, population dispersal • Legislation and protocols: Wildlife and Countryside Act, designated areas (SSSI to MCZ), CITES Appendices, IWC, EU CFP, ITTO - successes and limits of enforcement • Captive breeding and release: selection, difficulties, techniques, release site criteria, hard / soft release, monitoring; seed banks • Habitat conservation: creation, design (area, shape, corridors, age structure, colonisation), abiotic and biotic management, plagioclimax maintenance, population control, reintroductions • Habitat case studies (woodland, rainforest, coral reefs, islands, mangroves, Antarctica) illustrating threats and conservation efforts • Monitoring and new technologies informing decisions • Evaluation: examples of success (Iberian lynx, Hawaiian goose, Scilly rats, Yellowstone) and failure (ITTO, tigers); cost, politics, climate change; judgement on the most effective combination Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 26-71; AQA spec 3.1.2.3.	25

Question	Marking guidance	Marks
11.2	Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Physical features: mass / gravity retaining an atmosphere and giving pressure for liquid water; distance from the Sun and the Goldilocks zone; speed of rotation reducing temperature extremes; tilt producing seasons; magnetic field deflecting the solar wind; the Moon stabilising the orbit and producing tides • Importance of liquid water: solvent, transport, temperature control, anomalous expansion, high specific heat capacity, habitats, absorption of UV before the ozone layer • Development of life: Archaea at vents; photosynthesis; oxygen reacting with iron then accumulating; ozone layer allowing colonisation of land; aerobic organisms • Life changing conditions: carbon sequestration offsetting solar brightening, biogeochemical cycles, transpiration and rainfall, regulation of atmospheric gases • Methods of studying past conditions: proxy data, isotope ratios, ice cores, Mg:Ca ratios; understanding of continental drift and ocean currents • Links to current threats (climate change, ozone depletion) and a justified conclusion about the interdependence of physical and biological factors Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-9; AQA spec 3.1.1.	25